

MeshEcho: Confidence-Aware Route Selection for LoRa Mesh IoT

Zu Gao Zhi Quan
Shenzhen University, Shenzhen, China

Abstract—LoRa mesh IoT networks trade source-confirmed reliability against scarce airtime. A cached source route can remain weak when the channel changes. We compare MeshEcho’s original SNR-margin heuristic with one frozen correction that ranks paths by their weakest model-inferred hop PRR, without an extra hop penalty. In a packet-level simulator, 20 held-out topology seeds (51–70) yield 793 repeated-pair unicasts per policy under 6-dB block fading. The calibrated variant attains 0.624 ACK PDR versus 0.509 for old MeshEcho: a seed-paired +0.115 (95% CI [+0.031,+0.200]) at +2.47 s airtime and +5.09 J modeled energy per run. A PRR-product comparator with matched route-miss fallback attains 0.621 ACK PDR; the calibrated-minus-comparator interval crosses zero. Static-channel and isolated-first-discovery ACK gains over old MeshEcho are also uncertain; the latter has identical candidate sets for all 480 pairs. Reducing route TTL from 600 to 30 s under fading lowers calibrated ACK PDR by 0.304 and adds 41.05 s airtime. The result supports a bounded simulation correction, not strong-baseline superiority or hardware validity.

Index Terms—LoRa mesh, Internet of Things, route selection, fading, packet delivery ratio, airtime

I. INTRODUCTION

LoRa is attractive for low-power IoT because it combines long link budget with inexpensive radios [1]. A star topology is insufficient for obstructed or infrastructure-poor deployments, motivating multi-hop extensions [2]–[4]. Multi-hop forwarding is expensive: a low data rate occupies the channel for a long time, while a flood multiplies collision opportunities. A cached source route reduces redundancy, but one weak hop can invalidate an otherwise useful path.

MeshEcho records link observations during discovery and ranks candidate source routes. Its original score clips an SNR-margin heuristic and penalizes each additional hop. That combination can over-select a direct path that is temporarily favorable when the route remains cached across channel changes. We test one frozen correction: rank by the weakest model-inferred hop PRR from a received route request (RREQ), without a separate hop penalty. The default confidence threshold is zero, so a selected route does not trigger confidence-based flooding; bounded TTL-2 recovery occurs only when discovery finds no candidate. This is a simulation policy study, not a production mesh stack or an experimentally calibrated radio model.

This paper makes four contributions:

- a reproducible packet-level LoRa mesh harness with time-on-air, propagation, shadowing, half-duplex collision/capture, route discovery, and energy accounting;

- a prespecified 20-seed repeated-pair fading holdout, with matched PRR-product and route-miss-recovery controls and seed-paired intervals;
- an isolated 480-pair first-discovery comparison that fixes candidate exposure, plus a static-channel control; and
- short-TTL and ACK-feedback diagnostics that expose the airtime and energy costs of attempted stale-route recovery.

II. RELATED WORK

Prior ICC LoRa studies include analytical and simulation work on coverage, spreading-factor allocation, and frame error rates [5]–[9], as well as testbed validation [10], [11]. Multihop studies expose the flooding–airtime tradeoff [2], [4] and show that routing state and packet timing matter [12], [13]. ETX is a standard expected-transmission metric [14]; in our fixed-SF setting ETT ranks paths identically to ETX because packet time-on-air is constant. The stronger comparator here is the product of model-inferred hop PRRs, with a matched route-miss fallback control. We ask whether correcting MeshEcho’s heuristic improves ACK-confirmed delivery under fading without claiming superiority over that PRR-product baseline.

III. PROTOCOL AND MODEL

A. Network model

Nodes are uniformly placed in a square and share one half-duplex LoRa channel. Unless noted, packets have 32-byte payloads, 17 dBm transmit power, 125-kHz bandwidth, coding rate 4/5, capture threshold 6 dB, and SF7. Airtime follows the standard LoRa symbol calculation [15]. Log-distance path loss with log-normal shadowing produces SNR; reception is a logistic function of SNR margin, and overlapping packets are accepted only when the desired signal passes capture. The simulator records collisions, route-cache events, discovery success, and configured TX/RX energy.

B. Routing policies

Managed flooding rebroadcasts with duplicate suppression. Matched source-routing floods a request, collects candidates arriving in a 2-s window, and selects the shortest returned candidate. ETX and ETT minimize

$$C_{\text{ETX}} = \sum_i p_i^{-1}, \quad C_{\text{ETT}} = \sum_i \text{ToA}_i p_i^{-1},$$

where p_i is the simulator-predicted PRR from the SINR of a successfully received RREQ on hop i , not a PRR measured

from repeated packets. The min-hop baseline first minimizes hop count, then breaks ties by higher confidence score and lexicographic path order. Original MeshEcho selects the candidate with highest heuristic confidence; its calibrated variant maximizes the weakest modeled hop PRR. PRR-product maximizes the product of those same modeled PRRs. A matched PRR-product control also enables MeshEcho’s TTL-2 route-miss fallback; the plain PRR-product control does not. All routed policies use the same 2-s discovery window, 600-s route TTL unless stated, and no adaptive timeout retry. The historical Smart-CALM firmware/simulator namespace remains in the repository but is not an ICC protocol, ablation, or conclusion.

C. Confidence update and recovery

For each received route request, let m_i be the received SNR minus the SF-specific demodulation threshold on hop i , and let I_i indicate a reception marked as collided. The link score and MeshEcho’s accumulated candidate score are

$$\ell_i = \text{clip}_{.05}^{.98} \left(\frac{1}{1 + e^{-.7m_i}} - .18I_i \right), \quad (1)$$

$$c_1 = \ell_1, \quad c_i = \text{clip}_{.05}^{.98} (\min(c_{i-1}, \ell_i) - .025), \quad i \geq 2. \quad (2)$$

Here $\text{clip}_a^b(x) = \min(b, \max(a, x))$. The old score retains the weakest observed hop and charges 0.025 for each additional hop beyond the first. It is a heuristic, not an independently calibrated delivery probability. The frozen new variant uses the simulator’s same RREQ SINR-to-PRR mapping as ETX, but no 0.98 saturation or additional hop penalty:

$$\hat{p}_i = f_{\text{sim}}(\text{SINR}_i), \quad C_{\text{cal}} = \min_i \hat{p}_i, \quad C_{\text{prod}} = \prod_i \hat{p}_i. \quad (3)$$

Both scores use only successfully received RREQs; failed or censored requests do not supply an empirical PRR estimate. The destination retains every candidate arriving before the two-second window closes; the highest score wins, with shorter paths breaking equal-score ties. A route reply follows the reverse selected path, and data are sent only after the source receives that reply. Consequently, route-discovery success and ACK completion are distinct events, both included in the reported denominator.

For a cached route created at t_0 and expiring at t_e , forwarding uses

$$\hat{c}(t) = \text{clip}_0^1 \left[c_i - .1 \text{clip}_0^1 \left(\frac{t - t_0}{\max(1, t_e - t_0)} \right) \right]. \quad (4)$$

The default fallback threshold is zero, so this age reduction never launches a low-confidence duplicate for a selected route. If discovery observes no candidate, the pending unicast instead invokes a TTL-2 recovery flood. ACK-triggered cancellation applies to delayed fallbacks, but the evaluated configuration disables timeout retry. An optional ACK-guard variant evicts the same cached route after an unconfirmed source DATA transmission; it never retransmits that flow and cannot use future-channel knowledge. It is a diagnostic, not the proposed default.

TABLE I
FROZEN 2.1.24 REPEATED-PAIR HOLDOUT PARAMETERS.

Nodes / area	50 / 8250 m × 8250 m
Duration / load	600 s / 4 flows/min, 4 recurring pairs
Radio	32 B, 17 dBm, 125 kHz, CR 4/5, SF7
Path loss / shadowing	$n = 2.75 / \sigma = 4$ dB static
Fading / route TTL	6 dB per 60 s / 600 s (30 s control)
Discovery / pairs	2 s / 4, PRR ≥ 0.90, 2–4 hops
Seeds / unicasts	51–70 / 793 observed per policy

IV. EXPERIMENTAL DESIGN

A. Frozen split and repeated-pair workload

Seeds 41–50 were a development set for one score correction. After fixing the score, fallback controls, code, and comparisons, we ran untouched seeds 51–70 once as the holdout. The primary workload has 50 nodes in an 8250 m square, 600-s runs, four repeatedly selected directed pairs, SF7, 4-dB static shadowing, and 6-dB block fading every 60 s. Application unicasts follow a shared 4-flows/min Poisson trace; there are 793 scheduled and observed unicasts per policy across the 20 holdout seeds. No broadcast flows are scheduled. A shared analytical graph selects four pairs per seed with edge PRR ≥ 0.90 and graph distance 2–4 hops; the graph is never supplied to a routing policy. Every pair recurs in at least two fading blocks. The quality gate requires a complete pair pool, mean graph distance at least two, and at least 10% of direct links below 0.99 static PRR. These are conditioned repeated-pair estimates, not general random-pair performance.

B. Controls, candidate exposure, and uncertainty

The static control retains the same seeds, pairs, and application traces but sets temporal fading to zero. The short-TTL control retains 6-dB fading and reduces route TTL from 600 to 30 s. An optional 15-s ACK-guard variant is tested under the same traces; a later confirmed ACK cancels older guards for that exact route. All policies share the same topology, flow schedule, reception-stream construction, discovery window, and matched RREQ relay timing. Different forwarding decisions still change interference, so live candidate sets need not match. We compare their recorded path sets exactly.

The isolation check uses a fresh 12-s simulator for each of 24 directed pairs in each holdout seed and each of six policies: old MeshEcho, calibrated MeshEcho, ETX, PRR-product, min-hop, and matched source routing. It starts one unicast at 0.1 s, disables route-miss fallback, and uses the same node geometry and reception-stream construction. No preceding flow can alter a cache or collision history. The pair pool is graph-conditioned, but no policy receives that graph. The 2.1.23 one-shot seeds 21–40 are historical evidence and are not reused as a fresh holdout.

ACK PDR requires a source-received reverse-path ACK; destination PDR only requires DATA arrival. Each protocol’s seed-level PDR divides its ACK or destination count by that seed’s observed unicasts; the reported mean gives each of the 20 seeds equal weight. For a policy difference d_s , we report $\bar{d} \pm t_{.975, 19} s_d / \sqrt{20}$, pairing by seed. Isolation first averages

the 24 pair outcomes within each seed. Airtime and modeled TX/RX energy are per 600-s run, except isolation airtime, which is per pair. Flows within a seed are not independent statistical replicates.

V. RESULTS

A. Repeated-pair holdout

Table II reports seed means on the frozen 51–70 split. Under fading, the calibrated score increases ACK PDR from 0.509 to 0.624 relative to old MeshEcho, while consuming 2.5 s more airtime and 5.1 J more modeled energy per seed. It does not separate from the matched-fallback PRR-product control, which reaches 0.621 ACK PDR at 23.8 s airtime. The plain PRR-product row reaches 0.627/23.8 s but has a different route-miss recovery budget. Thus the old-score correction is supported; a new strong-baseline advantage is not.

The calibrated-minus-old fading ACK difference is +0.115 (paired 95% CI [+0.031, +0.200]), accompanied by +2.47 s airtime ([+1.23, +3.70]) and +5.09 J modeled energy ([+2.55, +7.63]). Its difference from PRR+fb is +0.003 ACK PDR ([−0.006, +0.011]). The latter uses the same fallback budget but remains a complete-policy comparison: among 83 paired discovery keys, candidate sets match in 81, and selected paths match in 78 of 80 both-selected events. Old and calibrated MeshEcho share only 51/83 candidate sets; within 48 equal-set/both-selected events, old MeshEcho chose a direct path 16 times and the calibrated variant zero times. Changed ranking is plausible, but different later interference prevents a per-flow causal attribution.

B. Candidate-controlled first discovery

The isolated experiment has exactly matching candidate sets for all 480 pairs across six policies; 458 pairs expose multiple routes. Table IV reports descriptive means. Calibrated minus old MeshEcho is +0.021 ACK PDR ([−0.013, +0.055]) at +0.034 s per pair ([+0.022, +0.046]). Calibrated minus PRR-product is +0.002 ACK PDR ([−0.006, +0.010]). Neither ACK interval excludes zero. Of 480 pairs, calibrated and PRR-product select the same path in 453; for all same-path policy pairs, ACK, destination, and airtime outcomes match. This supports route-choice attribution within the simulator, not a distinct calibrated advantage over PRR-product.

C. Controls and adverse results

Under the static control, calibrated and old MeshEcho reach 0.717 and 0.716 ACK PDR, a paired +0.000 difference ([−0.074, +0.075]). Calibrated airtime rises +1.22 s ([+0.09, +2.34]), and modeled energy rises +2.50 J ([+0.19, +4.82]). The fading improvement therefore does not generalize as a clear static gain. Under the 30-s TTL fading control, calibrated reaches 0.320 ACK PDR at 64.8 s airtime and 126.4 J energy. Against its own 600-s TTL result on identical seed-level traces, the paired ACK change is −0.304 ([−0.393, −0.215]), airtime changes by +41.05 s ([+31.40, +50.70]), and energy by +79.25 J ([+60.50, +98.01]). Route discoveries rise by 12.45 per seed.

This is a system-level TTL comparison, not an isolated cache mechanism effect.

ACK-guard eviction also fails as a recovery strategy: in the fading holdout, old MeshEcho’s ACK PDR falls from 0.509 to 0.471 while airtime rises from 21.2 to 41.1 s; PRR-product falls from 0.627 to 0.526 while airtime rises from 23.8 to 33.7 s. The fixed implementation cancels older pending guards when a later flow on the same route is acknowledged. It cannot infer from an absent ACK whether the DATA failed, the reverse ACK failed, or the path later recovered. These rows are negative diagnostics, not a proposed default.

Table V reports descriptive fading diagnostics omitted from the delivery table. Calibrated MeshEcho and PRR+fb both attempt 4.15 discoveries per seed; their mean ACK delays are 0.540 and 0.541 s among ACKed flows. ACK eviction raises discovery attempts, collision rate, and conditional delay while reducing cache hits. The legacy repair counter is 0.15 per seed for each non-eviction policy, but mixes cache expiry and failed discovery; it does not count successful route repairs.

The previous 2.1.23 one-shot and generalization results are distinct from this holdout. In an unconditioned random-pair case, managed flooding reached 0.736 ACK PDR/39.7 s versus old MeshEcho’s 0.430/103.2 s. A 100-node historical case selected only three-hop shortest graph distances and changed other settings; neither result validates the new score across arbitrary topologies. All airtime totals include discovery, DATA, ACK, and recovery transmissions. Destination delivery without a returned ACK is reported separately from ACK-confirmed completion.

VI. DISCUSSION AND LIMITATIONS

The calibrated score corrects a fading weakness of the original heuristic in the frozen repeated-pair workload, but it is a familiar bottleneck PRR metric, not evidence of a new routing optimum. Its ACK result is statistically indistinguishable from matched-fallback PRR-product, and its static and fixed-candidate ACK gains over old MeshEcho are uncertain. The fading cost is positive in airtime and modeled energy. The negative ACK-guard and short-TTL results rule out a simple stale-route repair claim.

All results are packet-level simulation. The model includes path loss, shadowing, collision timing, capture, route discovery, and energy accounting, but omits waveform interference, oscillator offset, modem implementation timing, duty-cycle enforcement, variable route headers, and antenna effects. The confidence estimate and forwarding channel share the same realization in static runs; temporal block fading is a deterministic stress model, not hardware validation. A hardware or hardware-in-the-loop study remains necessary for radio-specific external validity.

A. Threats to interpretation

The quality gate is a measurement contract, not a guarantee of deployment realism. Graph-qualified pair selection excludes disconnected sources and conditions the estimate on usable static links; the four-pair trace also favors cache reuse. The

TABLE II

FROZEN 51–70 REPEATED-PAIR HOLDOUT (20 SEEDS, 793 UNICASTS PER POLICY). ENTRIES ARE SEED MEANS. “CAL.” IS MODEL-DERIVED BOTTLENECK PRR; “PRR+FB” MATCHES THE TTL-2 ROUTE-MISS FALLBACK. ENERGY IS MODELED TX/RX CONSUMPTION PER 600-S RUN. THE FINAL TWO COLUMNS USE THE SAME SEEDS AND APPLICATION TRACES UNDER STATIC CHANNEL AND 30-S TTL FADING CONTROLS.

Policy	Fading ACK	Fading dest.	Airtime (s)	Energy (J)	Static ACK	TTL30 ACK
Old MeshEcho	.509	.535	21.2	42.1	.716	.298
Cal. MeshEcho	.624	.633	23.7	47.2	.717	.320
PRR-product	.627	.638	23.8	47.3	.715	.308
PRR+fb	.621	.633	23.8	47.5	.727	.317
MeshEcho+ACK evict	.471	.495	41.1	80.7	.690	.290
PRR+ACK evict	.526	.538	33.7	66.6	.714	.300

TABLE III

ACK-PDR DIFFERENCES FROM THE FROZEN HOLDOUT, PAIRED BY SEED ($n = 20$). INTERVALS ARE TWO-SIDED 95% t CIs. “FB” DENOTES MATCHED TTL-2 ROUTE-MISS FALLBACK.

Contrast	Channel/TTL	Δ ACK	95% CI
Cal. — old	Fading/600 s	+0.115	[+.031, +.200]
Cal. — old	Static/600 s	+0.000	[−.074, +.075]
Cal. — old	Fading/30 s	+0.022	[−.023, +.066]
Cal. — PRR+fb	Fading/600 s	+0.003	[−.006, +.011]
Cal. — PRR+fb	Static/600 s	−.010	[−.032, +.011]

TABLE IV

51–70 ISOLATED FIRST DISCOVERY: MEANS OVER 480 PAIRS, WITH IDENTICAL CANDIDATE SETS FOR ALL SIX POLICIES. AIRTIME IS PER PAIR.

Policy	ACK PDR	Dest. PDR	Airtime (s)
Matched source-route	.429	.498	3.766
Min-hop	.452	.512	3.772
ETX	.546	.594	3.803
Old MeshEcho	.681	.708	3.886
Cal. MeshEcho	.702	.719	3.920
PRR-product	.700	.715	3.923

old-score random-pair result shows why these strata must not be pooled. Candidate exposure is identical in isolation, but live repeated-pair policies diverge after their transmissions change later collisions. Shared seeds and reception-stream construction are not event-by-event common random numbers. All intervals use 20 topology seeds, not 793 or 480 independent packets. The 30-s TTL comparison changes the whole subsequent discovery and interference history, so it does not isolate one cache-event mechanism.

B. Deployment boundary

The modeled hop PRR comes from the SINR of a successful RREQ. It does not account for censored requests or calibrate itself against measured packet loss. Static shadowing lets discovery and forwarding share a link realization; the 60-s block-fading process is a synthetic stressor, not a measured channel trace. Moving nodes, antenna changes, duty-cycle limits, and external interferers may reorder routes before a 600-s TTL expires. The higher rediscovery cost with 30-s TTL shows that more frequent route selection alone is not a mobility solution.

A radio or hardware-in-the-loop study should preserve the pair schedule and two-second discovery window while logging

each request candidate, selected path, per-hop RSSI/SNR, DATA arrival, reverse ACK, and airtime. An SX126x or SX127x testbed with enforceable duty-cycle limits could determine whether the score ordering persists under real timing and interference. Until then, the results are simulator comparisons, not deployment guarantees.

VII. REPRODUCIBILITY

Version 2.1.24 includes source, tests, seed-level CSVs, summary CSVs, candidate-path records, and reproduction commands in the holdout 51–70 result group. The repeated-pair cases use 51–70; isolated rows use the same topology seeds with 24 fresh pairs per seed. Raw CSVs preserve each seed’s denominator, application-trace hash, discovery candidate paths, and selected routes. The 30-s TTL is specified by the case configuration and report header; the raw CSV identifies the case but has no separate numeric TTL column. Earlier 21–40 data remain versioned as 2.1.23 history and are not pooled with the holdout. No human participants or personal data are involved. AI-assisted drafting and code review were used; the numerical claims here were independently recomputed from raw CSVs. Final disclosure and author metadata remain the submitting authors’ responsibility.

VIII. CONCLUSION

Across 20 frozen fading seeds and 793 repeated-pair unicasts per policy, calibrated MeshEcho improves ACK PDR over the old score by +0.115 ([+.031, +0.200]) at +2.47 s airtime, but does not outperform a matched-fallback PRR-product baseline. Static and isolated ACK intervals cross zero; shortening TTL sharply reduces reliability and raises cost. The supported contribution is a reproducible, bounded score correction and honest comparison under a packet-level model, not universal routing superiority or physical validation.

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TABLE V

SECONDARY FADING DIAGNOSTICS ON SEEDS 51–70 (DESCRIPTIVE MEANS OVER 20 SEEDS, PER 600-S RUN). MEAN ACK DELAY AND WITHIN-SEED P95 ACK DELAY CONDITION ON ACKNOWLEDGED FLOWS; COLLISION RATE IS FAILED COLLISION-MARKED RECEPTIONS DIVIDED BY ALL RECEPTION ATTEMPTS. “DISC.” LISTS ROUTE-DISCOVERY ATTEMPTS/SUCCESSES AS SEPARATE SEED MEANS, NOT A POOLED FRACTION. “ACK INVALID.” COUNTS ACK-GUARD ROUTE INVALIDATIONS. NO PAIRED INFERENCE IS IMPLIED BY THESE MARGINAL MEANS.

Policy	Mean ACK (s)	P95 ACK (s)	Collision rate	Disc. att/succ.	Cache hits	ACK invalid.
Old MeshEcho	.460	1.488	.526	4.15/3.05	27.30	0
Cal. MeshEcho	.540	1.798	.473	4.15/2.95	26.25	0
PRR-product	.521	1.729	.471	4.15/2.90	25.80	0
PRR+fb	.541	1.736	.470	4.15/2.95	26.25	0
MeshEcho+ACK evict	.837	2.408	.629	9.80/7.60	17.05	5.80
PRR+ACK evict	.744	2.443	.562	7.30/5.15	18.70	3.15

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